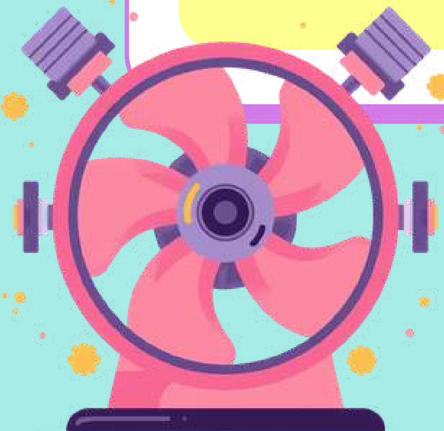
Planrs Academy

ELECTRIC MOTOR – HOW IT WORKS



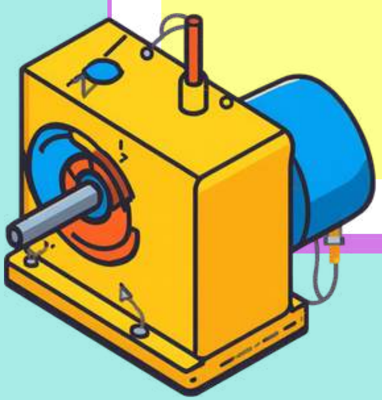
Can You Count How Many Things in Your Home Spin?

Look around your home! The ceiling fan, the mixer-grinder (mixi), the washing machine, even your toy car — can you guess what they all have in common?
They ALL share one secret — the Electric Motor!

What Is an Electric Motor?

An electric motor is a device that converts electrical energy into movement (mechanical energy). Think of it like magic — electricity goes in, and spinning motion comes out!

Electricity → Motor → Spin!



Spot the Motor Activity

Ceiling Fan ✓

Book ✗

Mixer-Grinder ✓

Pencil ✗

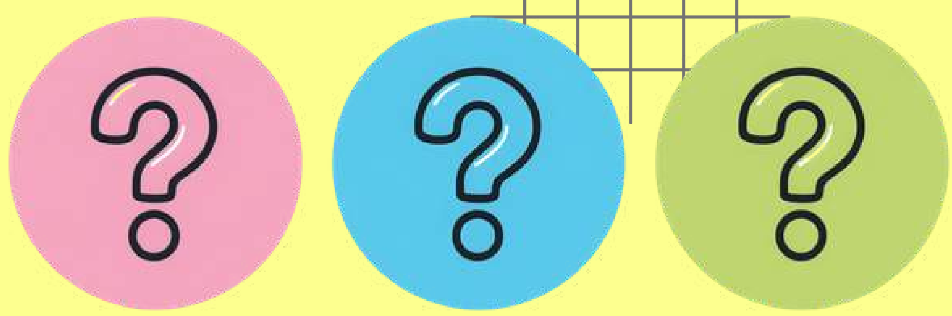
Metro Train ✓

The Secret Ingredients

Every motor needs just 3 things to work its magic...

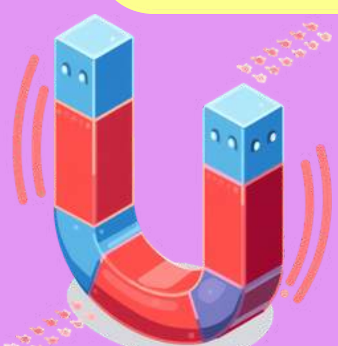
❓ Part 1 ❓ Part 2 ❓ Part 3

Can you guess what they are? Let's find out!



Part I: The Magnet

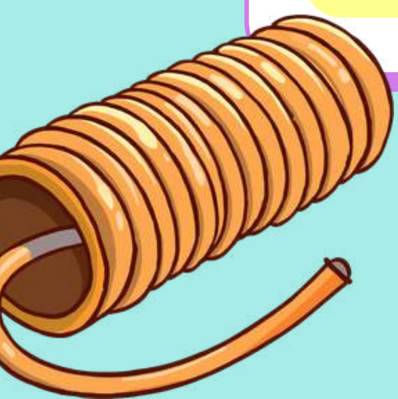
A MAGNET — the pusher & puller!
Magnets can push and pull things from a distance without touching them. You've seen this magic at home!
Think about fridge magnets holding your report cards on the refrigerator door.
Remember: Like poles REPEL (push away), unlike poles ATTRACT (pull together)!



Part 2: The Coil of Wire

THE SPINNER (ARMATURE)

A coil is a long wire wound many times into loops.
When electricity passes through the coil, it becomes an ELECTROMAGNET — a temporary magnet!
Think of it like winding a rubber band around a pencil many times.
The more loops, the stronger the magnet!



Part 3: Electricity

THE POWER SOURCE

ELECTRICITY is the energy that makes everything work! It comes from a battery or a plug point. When electricity flows through the coil of wire, it creates an electromagnet. This is the magic ingredient that powers the motor!

Three Parts Together

THE MOTOR FORMULA

When we combine all three parts, magic happens!
🧲 MAGNET — pushes and pulls the coil
🔄 COIL OF WIRE — spins when electricity flows
🔋 POWER SOURCE — gives energy to start it all
Remember: Magnet + Coil + Power = Motor! (M-C-P)

How Does It Work?

Let's see the magic happen! Get ready to discover the 4-step process that makes every motor spin. From ceiling fans to toy cars, the same magical steps power them all!

The 4 Steps of Motor Operation

Step 1: Switch ON → Electricity flows into the wire coil

Step 2: Coil becomes an ELECTROMAGNET (temporary magnet)

Step 3: Electromagnet is ATTRACTED & REPELLED → Coil SPINS

Step 4: Shaft spins → Fan blade or wheel turns!

The cycle repeats continuously!

Real-Life Connections — Motor Safari

Ceiling Fan — spins ~400 times per minute!

Mixer-Grinder (Mixi) — spins ~18,000 times per minute!

Delhi Metro — traction motors power the entire train

E-Auto & Ola Electric Scooter — motors on the go!

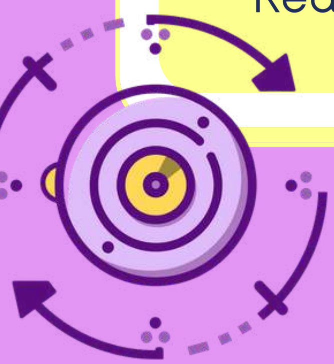
Diwali Spinning Light Toys — festive motor magic!

More Motor Examples

- Water Pump (Motor Pump)
- Electric Pencil Sharpener
- Battery-Operated Toy Cars
- Washing Machines
- Exhaust Fans

Let's Be the Motor!

Stand up and become a motor! Stretch your arms out wide — you are now a coil of wire. Spin slowly for low electricity. Spin faster for more electricity! Stop when the switch is OFF. Ready? Let's go!



Quick Recap

3 Main Parts: Magnet, Coil of Wire, Power Source

4 Steps: Power ON → Coil becomes Electromagnet → Spin → Movement!

Energy Conversion: Electrical Energy → Mechanical Energy

Remember: M-C-P = Motor!

Electricity makes things SPIN!

Check Your Understanding

Q1: Name 2 things at home with a motor

Q2: What 3 parts does a motor need?

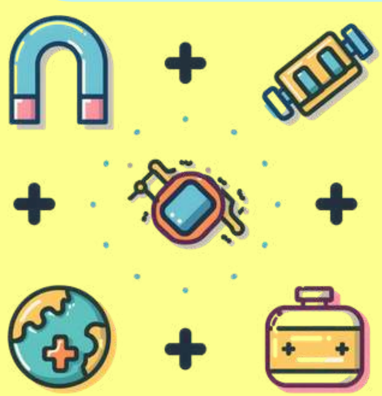
Q3: What energy goes IN the motor?

Q4: What comes OUT of the motor?

Q5: True or False: A book uses a motor

Today's Big Idea

⭐ ONE BIG IDEA: A motor turns electricity into spinning movement!
⭐ THREE PARTS: Magnet + Coil + Power = Electric Motor
⭐ ONE STORY: Every fan, mixi, and toy car spins because of electricity and magnetism working together!



Homework & Closing

Motor Detective Mission!
Go home and find ALL motors you can spot.
Make a list!
Draw one motor and label its parts.
Next time: How does a generator work?
Great job today! Keep exploring!
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THANK YOU!

KEEP EXPLORING AND STAY CURIOUS!